

Arduino-Based Smart Security System Using Ultrasonic Sensor

Project Report

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Abstract

This project presents an Arduino-based Smart Security System that uses an ultrasonic sensor to detect nearby objects. The ultrasonic sensor continuously measures the distance between the sensor and an object. When an object comes within the predefined range, the Arduino processes the sensor data and activates an LED and a piezoelectric buzzer as warning indicators. When the object moves away, the LED and buzzer automatically turn off.

The project demonstrates the basic concepts of distance measurement, sensor interfacing, and automation using Arduino Uno. It is simple, cost-effective, and easy to build, making it suitable for learning embedded systems and electronics. Such a system can be used for basic security applications in homes, offices, laboratories, and restricted areas where intrusion detection is required.

Problem Statement

Ensuring the security of homes, offices, laboratories, and other restricted areas is an important challenge. In many cases, continuous human monitoring is not possible, and advanced security systems can be expensive and difficult to install. As a result, there is a need for a simple, affordable, and reliable system that can automatically detect the presence of an object or an intruder and immediately alert the user.

This project addresses this problem by developing an Arduino-based Smart Security System using an ultrasonic sensor. The sensor continuously measures the distance of nearby objects, while the Arduino processes the data in real time. If an object is detected within a predefined range, the system activates an LED and a buzzer to provide an instant visual and audible alert. This project demonstrates how embedded systems can be used to develop an effective and low-cost security solution for basic monitoring applications.

Objectives

The main objectives of this project are:

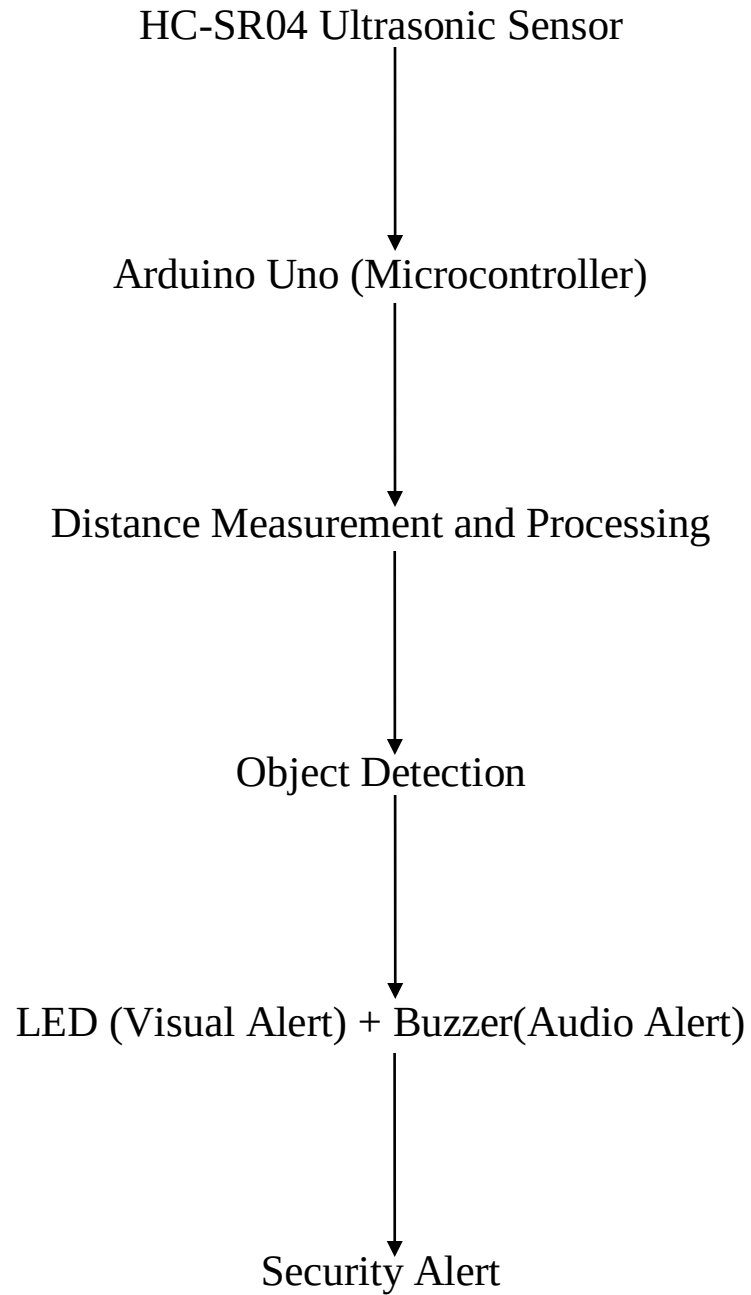
- To design and develop a smart security system using the Arduino Uno and an ultrasonic sensor.
- To detect the presence of an object or intruder within a predefined distance.
- To provide an immediate visual and audible alert using an LED and a buzzer when an object is detected.
- To demonstrate the integration of sensors, actuators, and a microcontroller in an embedded system.
- To develop a simple, low-cost, and reliable security solution for basic monitoring applications.
- To understand the practical implementation of distance measurement, real-time sensing, and Arduino programming.

Components Used

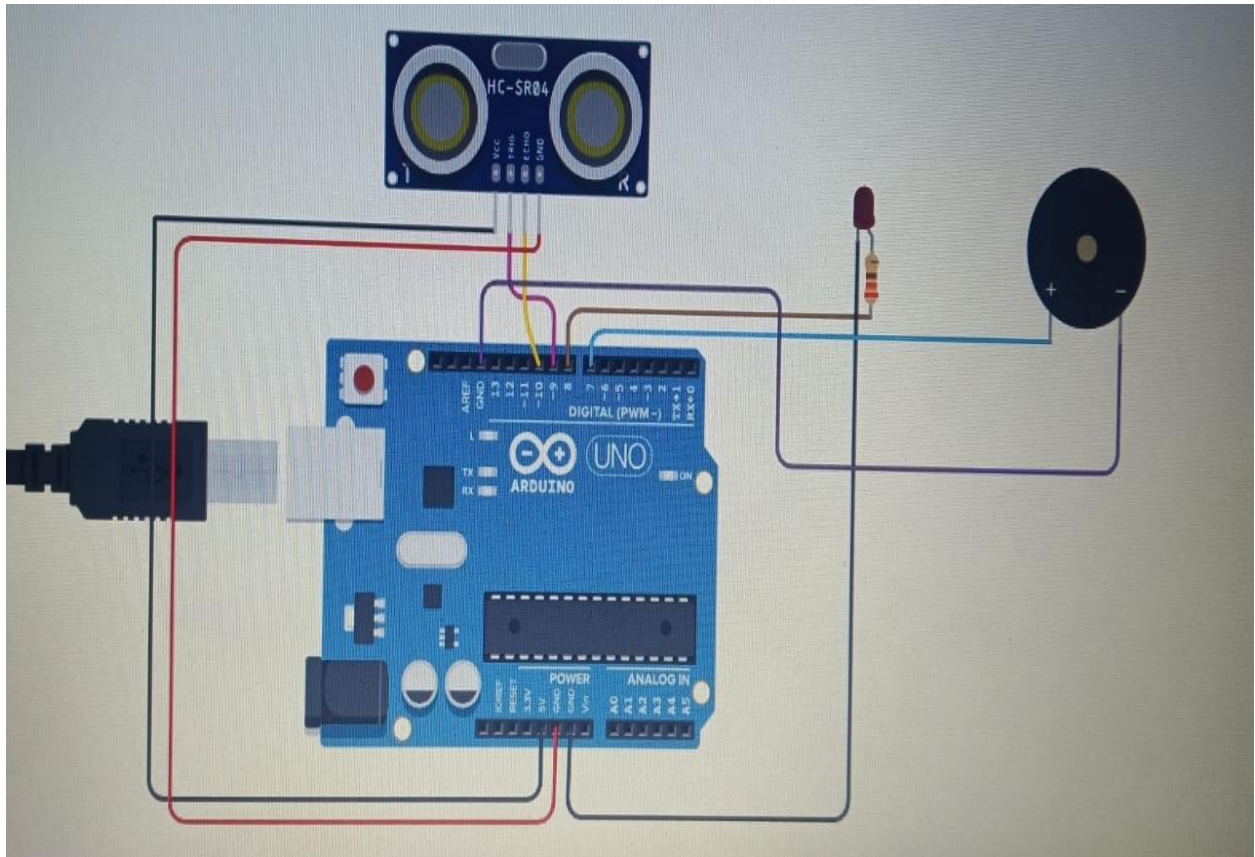
The following hardware components were used to develop the Arduino-based Smart Security System:

Component	Quantity	Purpose
Arduino Uno	1	Controls the entire system and processes sensor data.
HC-SR04 Ultrasonic Sensor	1	Detects objects by measuring the distance using ultrasonic waves.
Piezoelectric Buzzer	1	Produces an audible alarm when an object is detected.
LED	1	Provides a visual indication of object detection.
220 Ω Resistor	1	Limits the current flowing through the LED to protect it.
Breadboard	1	Used for assembling the circuit without soldering.
Jumper Wires	As required.	Used to make electrical connections between components.
USB Cable	1	Powers the Arduino and uploads the program from the computer.

Block Diagram:



Circuit Diagram:



Working Principle

The Arduino-based Smart Security System works by continuously monitoring the area using an HC-SR04 ultrasonic sensor. The sensor sends out ultrasonic sound waves and measures the time taken for the echo to return after hitting an object. Using this time, the Arduino calculates the distance between the sensor and the object.

When an object or person comes within the predefined distance (for example, less than 20 cm), the Arduino detects it as a possible intrusion. It then activates the connected output devices, such as a buzzer to produce an alarm and a DC motor (if used) to simulate opening/closing a security gate or rotating a surveillance mechanism.

If no object is detected within the set distance, the system remains in monitoring mode. This process repeats continuously, allowing the system to detect intruders in real time.

Flow of operation:

1. Power ON the Arduino.
2. Ultrasonic sensor transmits ultrasonic waves.
3. Echo is received from nearby objects.
4. Arduino calculates the distance.
5. If the distance is below the threshold:
6. Buzzer turns ON.
7. DC motor activates (if connected).
8. If the distance is above the threshold:
9. Buzzer remains OFF.
10. Motor remains OFF.
11. The system continuously repeats the detection process.

Program:

```
// Arduino Smart Security System using HC-SR04
```

```
const int trigPin = 9;
```

```
const int echoPin = 10;
```

```
const int buzzer = 8;
```

```
const int led = 7;
```

```
long duration;
```

```
int distance;
```

```
void setup() {
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  pinMode(buzzer, OUTPUT);
```

```
  pinMode(led, OUTPUT);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
  // Send ultrasonic pulse
```

```
  digitalWrite(trigPin, LOW);
```

```
  delayMicroseconds(2);
```

```
  digitalWrite(trigPin, HIGH);
```

```
  delayMicroseconds(10);
```

```
  digitalWrite(trigPin, LOW);
```

```
  // Read echo
```

```
  duration = pulseIn(echoPin, HIGH);
```

```
  // Calculate distance (cm)
```

```
  distance = duration * 0.034 / 2;
```

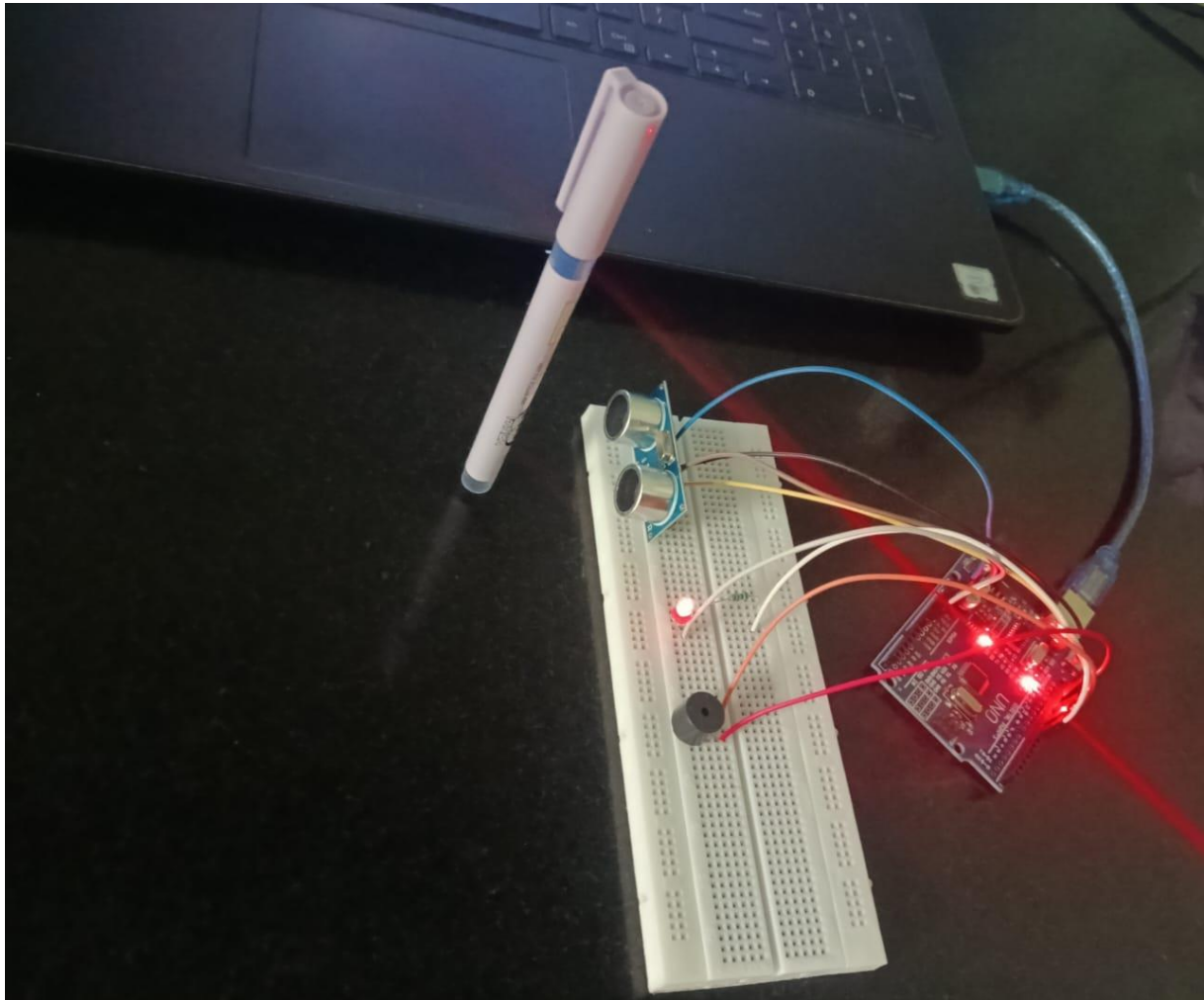
```
  Serial.print("Distance: ");
```

```
Serial.print(distance);  
Serial.println(" cm");  
  
// Security condition  
if (distance > 0 && distance <= 20) {  
    digitalWrite(led, HIGH);  
    digitalWrite(buzzer, HIGH);  
} else {  
    digitalWrite(led, LOW);  
    digitalWrite(buzzer, LOW);  
}  
  
delay(200);  
}
```

Result

The Arduino-Based Smart Security System was successfully designed, implemented, and tested. The HC-SR04 ultrasonic sensor accurately detected objects entering the predefined range. When an object was detected within the set distance, the Arduino activated the buzzer to generate an alert and the DC motor operated as programmed. When no object was present within the threshold distance, the system remained inactive. The project demonstrated reliable intrusion detection and real-time response, confirming that the system can be used as a simple and cost-effective security solution for homes, offices, and other restricted areas.

Testing:



Applications

- Home security systems to detect unauthorized entry.
- Office and commercial building security.
- Restricted areas such as laboratories, server rooms, and storage rooms.
- Banks and lockers for basic intrusion detection.
- Warehouses and industrial facilities for monitoring entry points.
- Museums and exhibition halls to protect valuable items.
- Parking areas and garages for vehicle detection.
- Educational projects for learning embedded systems, sensors, and Arduino programming.

Advantages

- Low-cost and economical security solution.
- Provides real-time intrusion detection.
- Easy to design, install, and maintain.
- Low power consumption, making it suitable for continuous operation.
- Compact and portable system.
- Can be expanded by integrating additional sensors, cameras, GSM, Wi-Fi, or IoT modules.
- Improves safety by providing an immediate alert when an intruder is detected.
- Suitable for educational, residential, and small commercial security applications.

Limitations

- The system detects objects only within the range of the ultrasonic sensor.
- Performance may be affected by obstacles, irregular surfaces, or environmental conditions.
- It cannot identify or distinguish between different objects or individuals.
- The system provides only local alerts and does not send notifications to a mobile phone or remote user.
- Continuous power supply is required for uninterrupted operation.
- It is suitable for small-scale security applications and may require additional sensors for monitoring larger areas.

Future Scope

- Integrate IoT technology to send real-time alerts and notifications to a smartphone.
- Add a camera module for image or video capture when an intrusion is detected.
- Implement face recognition or artificial intelligence for identifying authorized and unauthorized persons.
- Enhance the system by integrating GSM or Wi-Fi modules for remote monitoring and control.
- Use multiple sensors to provide wider area coverage and improve detection accuracy.
- Include battery backup to ensure uninterrupted operation during power failures.
- Develop a mobile or web application for monitoring the security system from any location.

Conclusion

The Arduino-Based Smart Security System was successfully designed and tested using an Arduino Uno and an HC-SR04 ultrasonic sensor. The system effectively detected nearby objects and activated the buzzer and DC motor whenever an object entered the predefined range. The project demonstrates a simple, reliable, and cost-effective approach to intrusion detection. It also provides a good foundation for developing more advanced security systems by integrating wireless communication, IoT, and smart monitoring technologies.

References

1. Arduino Documentation. Arduino Uno Rev3. <https://docs.arduino.cc/>
2. HC-SR04 Ultrasonic Sensor Datasheet.
3. Arduino IDE Documentation. <https://www.arduino.cc/en/software>
4. Autodesk. Tinkercad Circuits. <https://www.tinkercad.com/>